



User Behaviour OCA Indonesia

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Executive Summary

Project Description :

This project analyzes active user behavior on OCA Indonesia's CPaaS platform using multi-channel transaction data. The analysis focuses on understanding user activity patterns, usage intensity, and engagement recency across WhatsApp, SMS, Email, and Call channels. Users are segmented using the RFM (Recency, Frequency, Monetary) framework to identify super users, loyal clients, at-risk users, and upsell opportunities.

Objectives :

- Identify different types of active users based on usage behavior
- Detect early signals of churn and inactivity
- Highlight high-value users and super users
- Support personalization, retention, and upsell strategies
- Provide actionable segmentation for Product and Account teams

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Executive Summary

Tools :

- **EDA** : Big Query, Spreadsheet
- **Visualisation** : Tableau

Metrics :

- RFM : the data only contain 20 users, with 3 super users.
 - Channel diversity
 - Active days
 - Most active hour / day
 - Delivery success rate
- } explain behavioral differences and translate segments into actionable business strategies

Analysis Result:

- Revenue is highly concentrated into 3 super users.
- Monetization is channel-driven, not volume-driven, with WhatsApp and SMS as the primary revenue drivers while Email drives high volume with low value.
- Usage indicates strong product stickiness with clear growth tiers.
- Recency Is Not Yet a Differentiator.

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Business Background

OCA Indonesia is a CPaaS provider under Telkom Indonesia that enables businesses to send large-scale messages across WhatsApp, SMS, Email, and Voice channels, where maintaining delivery reliability, billing accuracy, and cross-channel visibility is critical to sustaining revenue and customer trust.

Given the limited observation window of three months and a small user base of 20 clients, this analysis focuses on descriptive behavioral patterns and early signals rather than long-term churn prediction.

Note : “This project is part of the RevoU x Telkom Indonesia Virtual Internship Program.”

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Business Problem

OCA currently lacks a clear understanding of how active users differ in terms of engagement, value, and churn risk. Without user-level segmentation, the company risks applying generic strategies to all clients, leading to inefficient retention efforts, missed upsell opportunities, and delayed churn detection.

Objective of analysis

To segment users based on recency, frequency, and monetary value in order to prioritize retention, personalize engagement strategies, and identify high-risk and high-potential users.

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Methodology

1

Problem Understanding

identify high-risk segments and optimize underwriting decisions.

2

Data Preparation & Data Cleaning

Update created_at and data timestamp to year 20XX from 00XX. Since it's a timestamp, UTC is default.
Conclusion : the time already Machine server timezone.

3

Exploratory Data Analysis

Conduct Initial Investigation on dataset to define metrics and test for hypothesis

4

Data Analysis

Conduct the analysis with the metrics that decided on EDA with SQL and excel.

5

Data Visualization

Visualize the analysis into chart, table, and dashboard by using Tableau

6

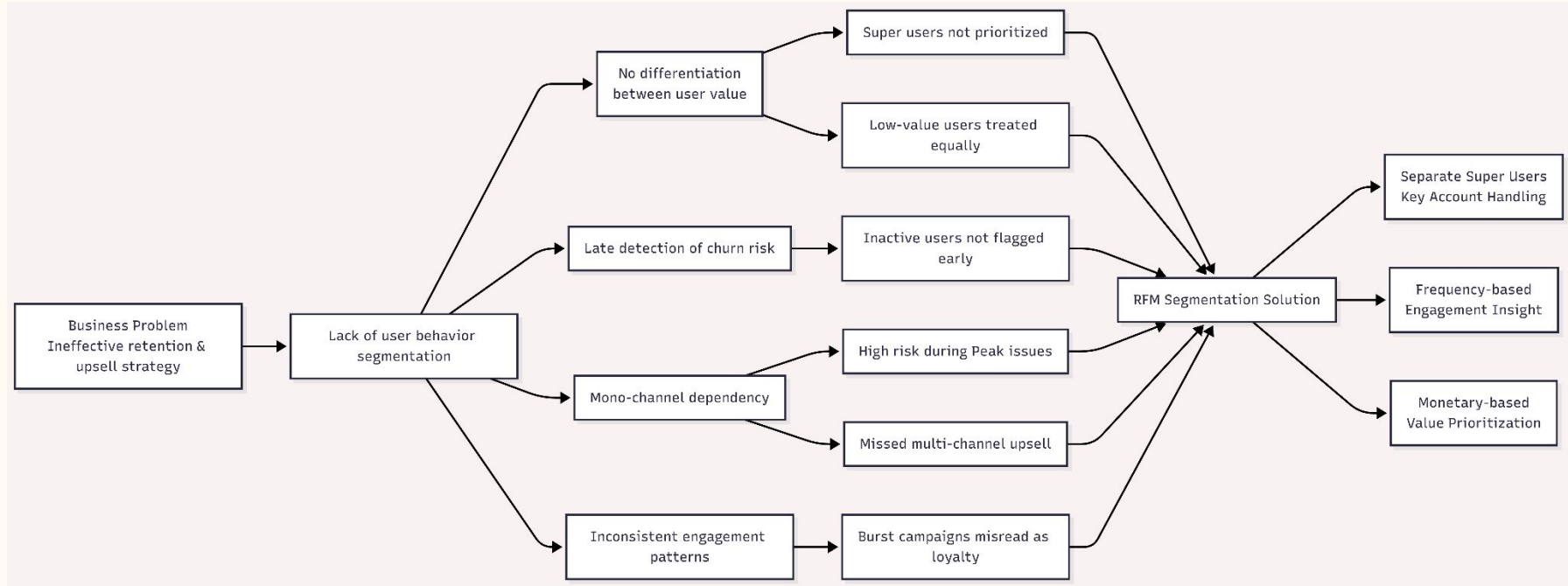
Insight & Recommendation

Use the result of data analysis to determined the insight and business recommendation to solve the problem

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Root Cause Analysis



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Dataset Scope

Table :

- **Call** : Is_Charge, Created_at, User_id, Total_Price
- **Sms** : Is_Charge, Created_ad, User_id, Total_sent, Total_Price
- **Whatsapp** : Is_Charge, Created_at, User_id, Price
- **Email** : Is_Charge, Created_at, User_id, Price
- **Users** : User_id, User_Name

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What is RFM?

RFM is a widely used business framework that helps us understand client behavior using three simple dimensions: how recently they used our platform (Recency), how frequently they send messages (Frequency), and how much billable usage they generate (Monetary).

Why RFM for OCA (CPaaS)

- OCA revenue is **usage-based**, not subscription-based
- Client value depends on **activity freshness, volume, and billable usage**
- RFM enables:
 - Early **churn detection**
 - **Account prioritization**
 - Targeted **upsell & retention strategies**

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Super User and RFM Segmentation Result

Super User (not included to RFM)

- CV Wisata Indo
- PT Dana Nusantara
- CV Land Sejahtera

In this dataset, all users have identical recency values within the analysis window, meaning recency does not currently differentiate churn risk. Therefore, recency was excluded as a dominant factor in churn classification and RFM was primarily used to represent relative user value rather than inactivity risk.

Potential Loyalist

- PT Travel Nusantara
- PT Saham Global
- PT Estate Sejahtera
- CV Jalan Mandiri
- PT Konektivitas Indo
- PT Karya Sejahtera
- PT Logistik Mandiri
- PT Retail Sejahtera
- PT Belajar Nusantara
- CV Niaga Indo
- CV Hotel Jaya

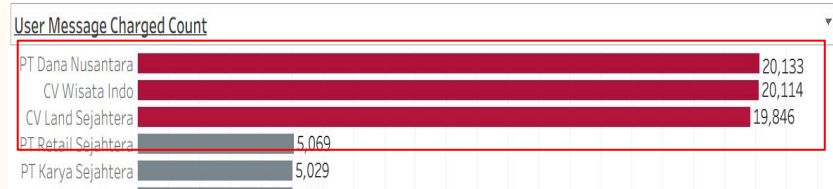
Regular User

- PT Mart Jaya
- CV Mart Indo
- CV Hotel Nusantara
- PT Edu Mandiri
- PT Karya Jaya
- PT Citra Global

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Super Users



3 users dominate monetary value and frequency :

- PT. Dana Nusantara
- CV Wisata Indo
- CV Land Sejahtera

Impact

- Revenue concentration risk

Recommendation :

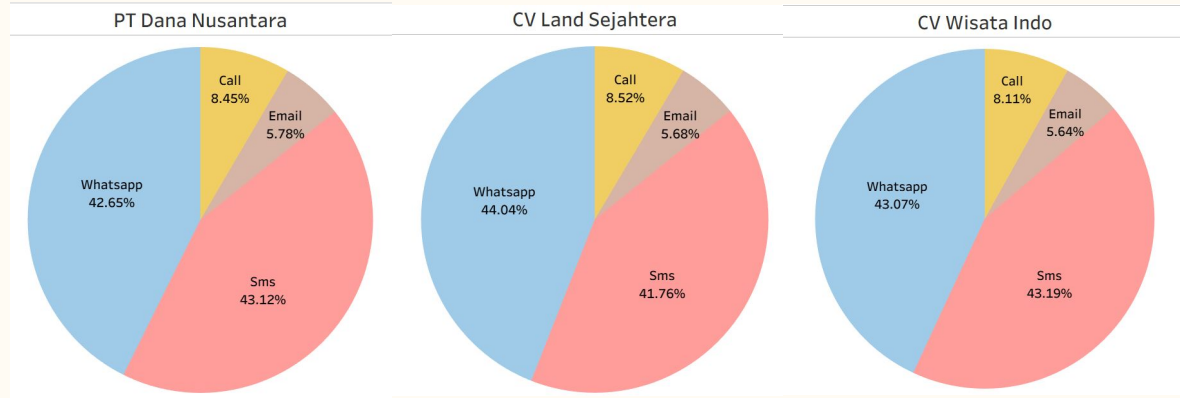
- 👉 Assign **dedicated account management**
- 👉 Proactive SLA & delivery monitoring
- 👉 Early renewal & upsell discussion

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Super Users - Total Price Charged

Total Price Charged



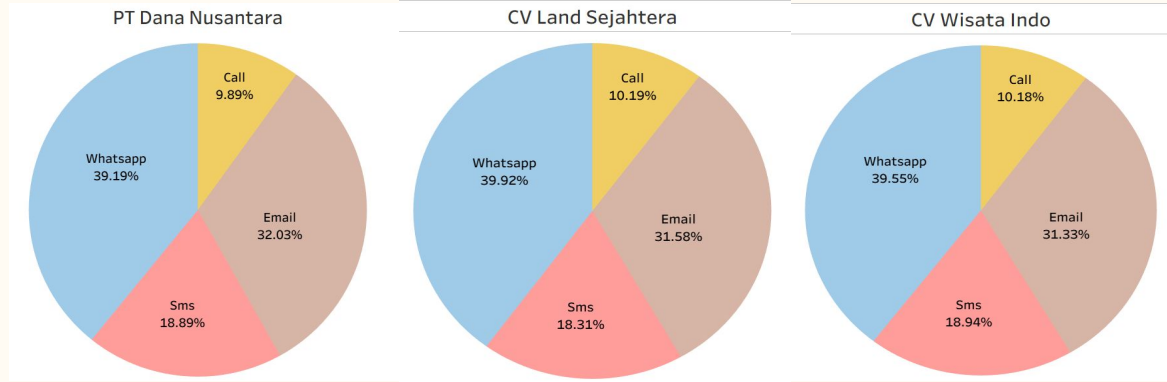
Across super users, channel usage patterns are largely consistent, with WhatsApp and SMS contributing the majority of platform usage. This indicates that super users primarily rely on these channels as their core communication tools, reflecting common operational requirements rather than user-specific channel preferences.

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Super Users - Total Message Sent

Total Message Sent



While WhatsApp and SMS dominate revenue contribution, message volume is driven mainly by WhatsApp and Email. This suggests that high message volume does not always translate directly into higher revenue, highlighting differences in channel pricing and usage intensity across communication channels.

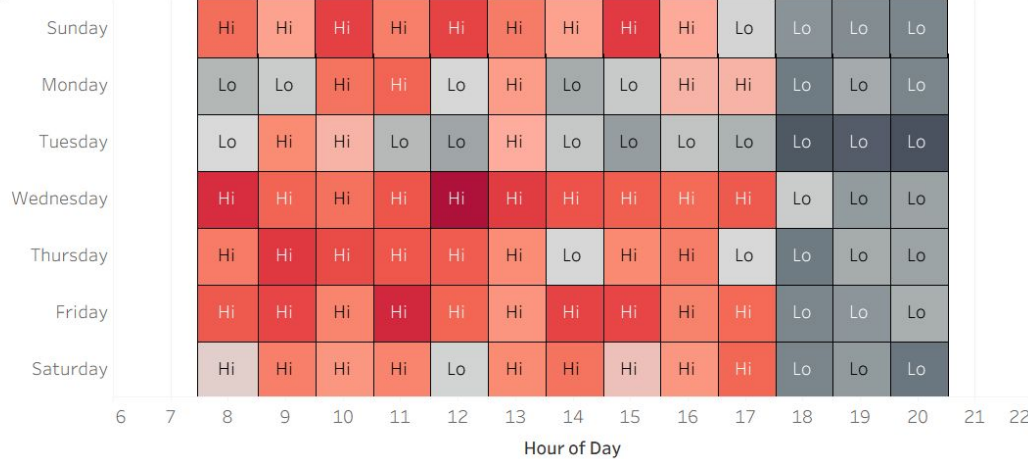
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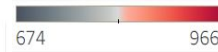
Super Users - Peak Days and Hours

Hour x Day Message Volume

Day of Week ▾



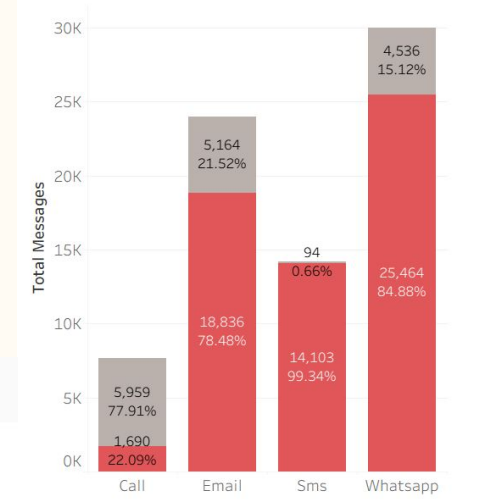
Total Message



Is Charge



Delivery Trend



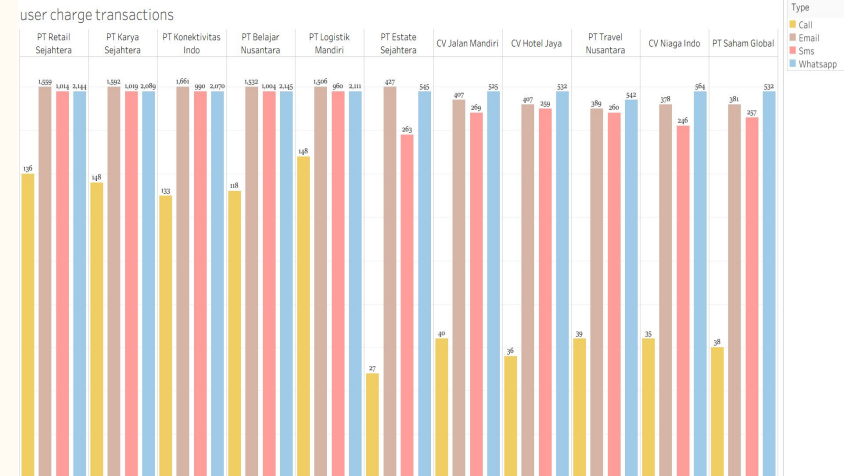
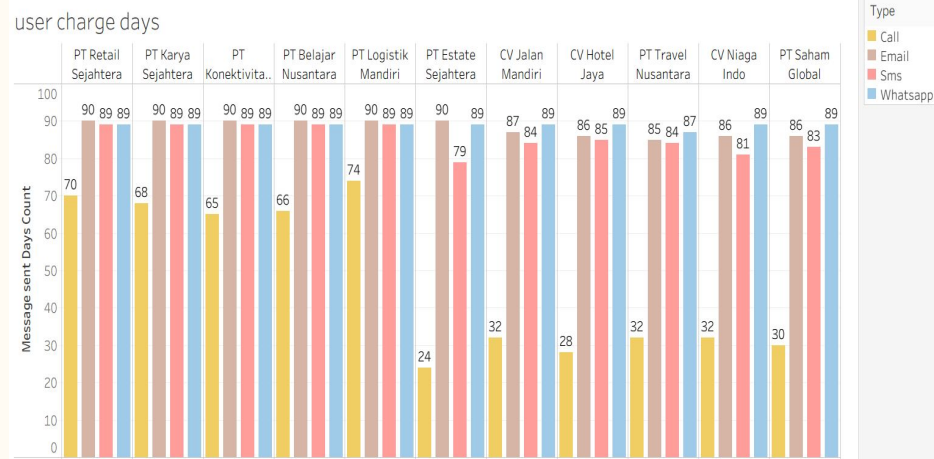
Engagement Peaks: Super Users peak on Wednesdays, with daily activity highest between 08:00 and 17:00.

Delivery Performance: Calls see the highest failure rate, while SMS remains the most reliable channel.

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Potential Loyalist User Charge



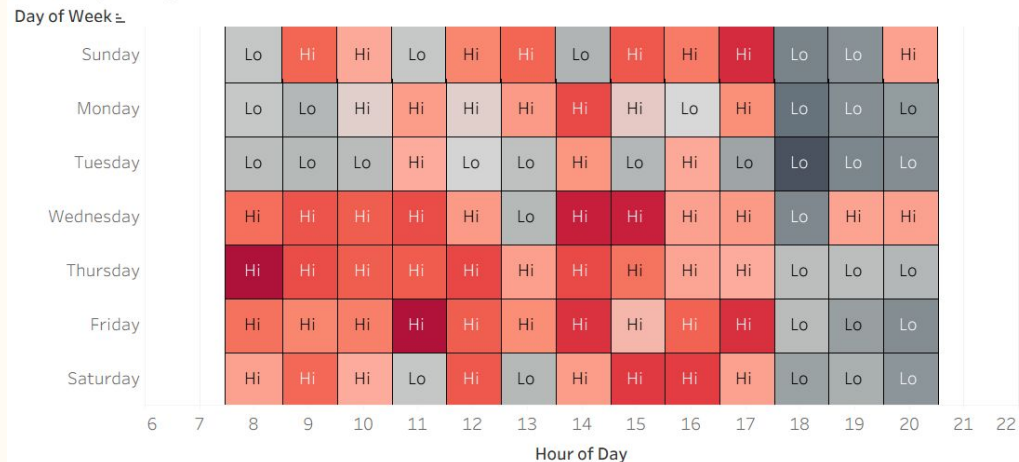
The chart shows that most users are consistently active across WhatsApp, SMS, and Email on the majority of days within the three-month period, indicating stable and recurring platform usage rather than short-term campaigns. In contrast, Call usage appears significantly lower and less consistent across users, suggesting that calls are used more selectively or for specific use cases. Overall, this pattern highlights WhatsApp and SMS as the core operational channels, while Email supports high-frequency messaging and Calls remain a supplementary channel.

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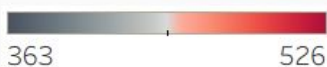


Potential Loyalist - Peak Days and Hours

Hour x Day Message Volume



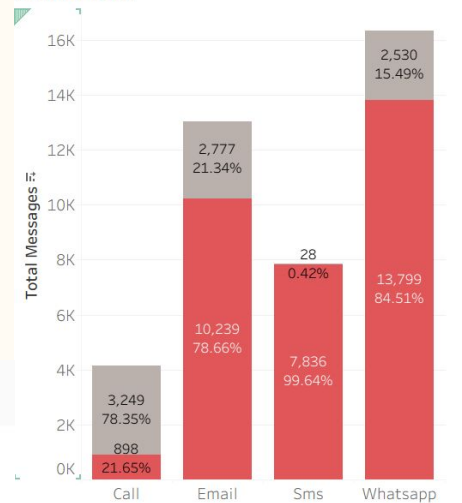
Total Message



Is Charge



Delivery Trend



Engagement Peaks

Peak Windows: Activity is concentrated midweek, with Wednesdays seeing sustained high volume and 08:00–17:00 acting as the daily peak (excluding Tuesdays).

Delivery Performance

Channel Mix: SMS and Email serve as the primary communication pillars.

Reliability: SMS delivers the highest value and success rate, whereas Voice/Call remains the primary driver of delivery failures.

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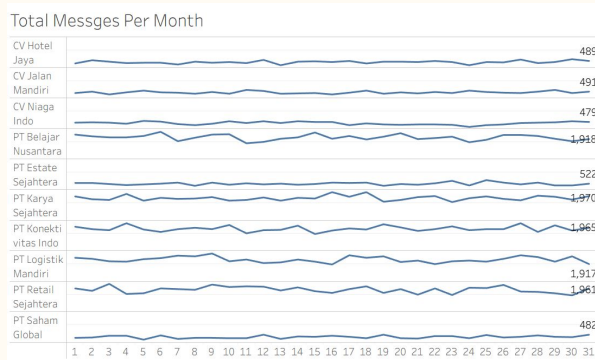


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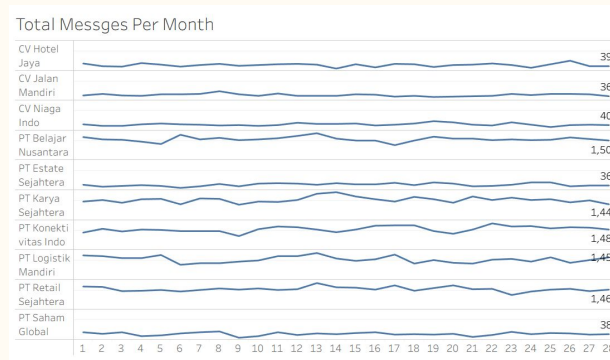


Potential Loyalist - monthly Growth

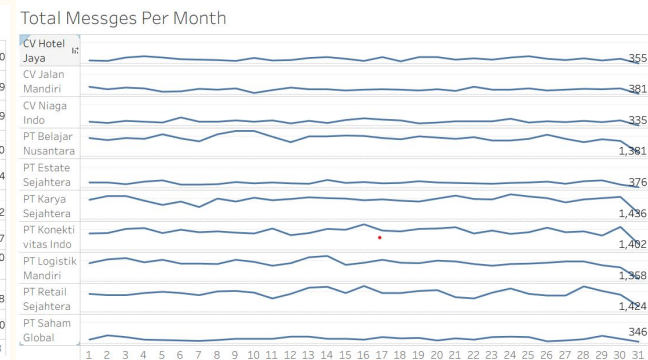
January :



February:



March :



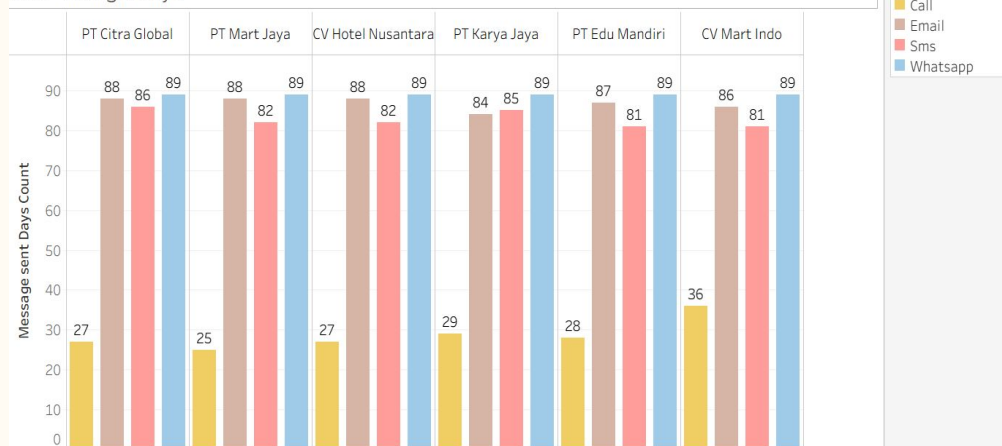
Message Count daily is **stable and recurring**, indicating embedded workflows rather than one-off campaigns. However, we have observed a marginal month-over-month decline in total message volume.

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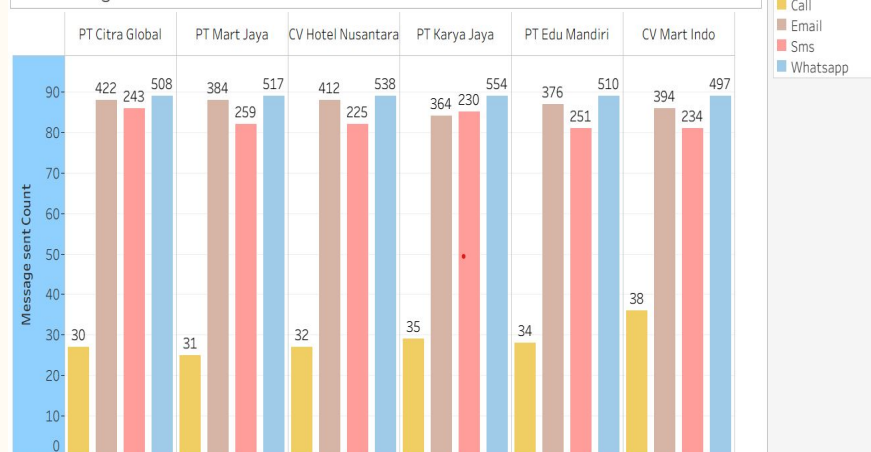


Regular User - User Charge

user charge days



user charge transactions



Regular Users demonstrate **lower and less consistent activity** across all channels.

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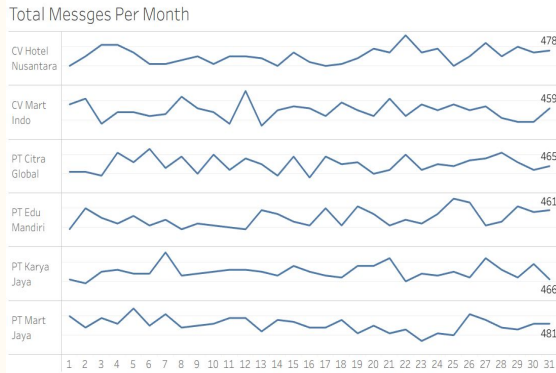


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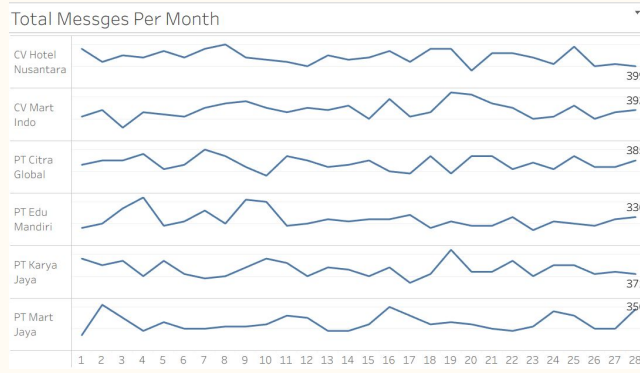


Regular User - Monthly Growth

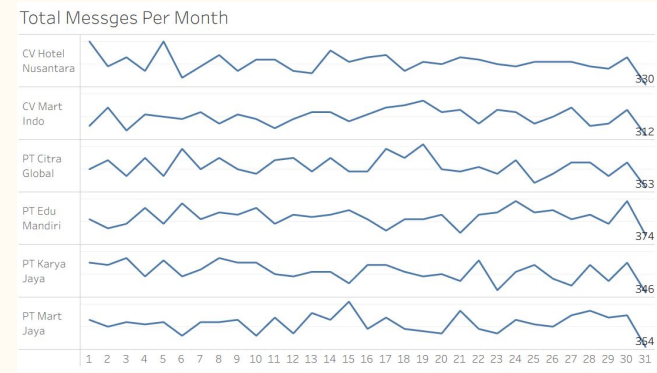
January :



February:



March :



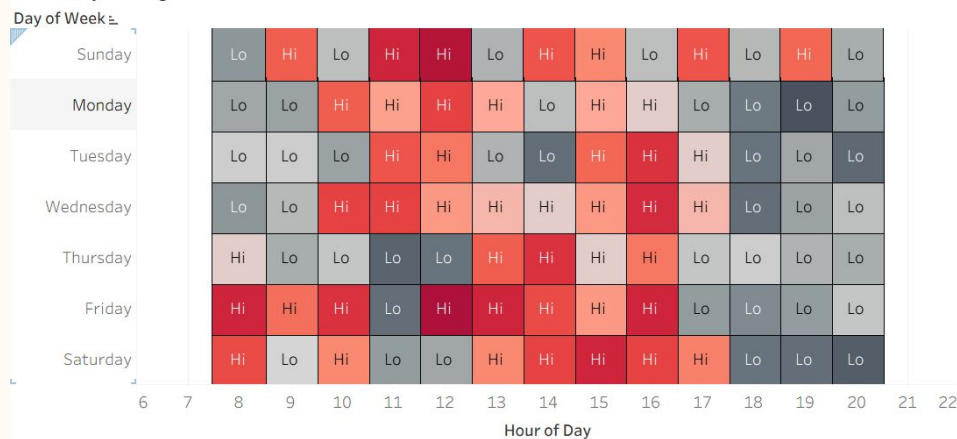
Month-to-month usage is **flat or declining**, with no clear growth trend.

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Regular User Peak Days and Hours

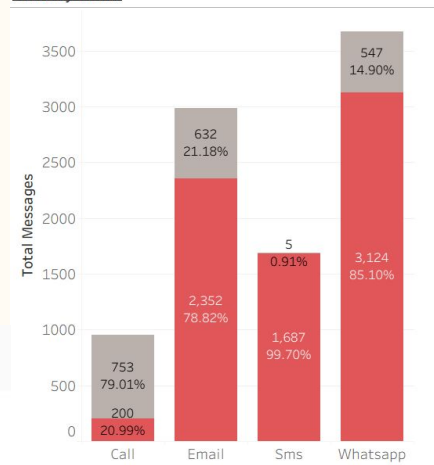
Hour x Day Message Volume



Is Charge

Failed
Charged

Delivery Trend



Engagement Peaks

- **Peak Hours:** Consistent daily window from 08:00–17:00.
- **Peak Days:** Inconsistent; no single day emerges as a primary driver for this segment.

Delivery Performance

- **Primary Channels:** Email and WhatsApp (Highest Volume).
- **Reliability:** SMS shows optimal success (Best Value), whereas Calls continue to struggle with high failure percentages.

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Insight And Recommendations

Super Users

- **Insight:** Extremely high revenue and frequency, concentrated on WhatsApp and SMS; Email drives volume, not value. Wednesday become peak day, while peak hours almost daily at 8.00 to 17.00.
- **Risk:** Revenue concentration and operational dependency.
- **Action:** Dedicated account management, proactive SLA monitoring, premium feature upsell.

Potential Loyalists

- **Insight:** Stable, recurring multi-channel usage with strong engagement but moderate spend.
- **Opportunity:** Highest upsell and expansion potential.
- **Action:** Usage-based incentives, channel optimization education, bundled pricing strategies.

Regular Users

- **Insight:** Low frequency, low spend, inconsistent engagement.
- **Risk:** Early churn or long-term inactivity.
- **Action:** Activation programs, lightweight onboarding, selective retention investment.

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[You can check the tableau public here](#)

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Appendix - RFM Python Code

Import Library

```
1 import pandas as pd
2 import numpy as np
3 from datetime import timedelta
```

Import Data

```
1 df = pd.read_csv("OCA.csv")
2 df.head()
```



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Appendix - RFM Python Code

```
1 df['created_at'] = pd.to_datetime(df['created_at'])
```

```
1 analysis_date = df['created_at'].max() + timedelta(days=1)
2 window_start = analysis_date - timedelta(days=90)
3
4 df_90 = df[df['created_at'] >= window_start]
5
```

```
1 rfm = df_90.groupby('user_id').agg(
2     recency=('created_at', lambda x: (analysis_date - x.max()).days),
3     frequency=('transaction_id', 'count'),
4     monetary=('total_price', 'sum')
5 ).reset_index()
6
7 rfm
8
```

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Appendix - RFM Python Code

Super User separate

```
1 super_users = (  
2     rfm.sort_values('monetary', ascending=False)  
3     .head(3)  
4     .assign(segment='Super User')  
5 )  
6  
7 rfm_regular = rfm[~rfm['user_id'].isin(super_users['user_id'])].copy()
```

created a copy of the filtered DataFrame to avoid chained assignment and ensure safe transformations during RFM scoring

RFM

```
1 rfm_regular
```

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Appendix - RFM Python Code

Recency

```
1 rfm_regular['R_score'] = pd.qcut(
2     rfm_regular['recency'].rank(method='first'),
3     3,
4     labels=[3,2,1]
5 ).astype(int)
```

Frequency

```
1 rfm_regular['F_score'] = pd.qcut(
2     rfm_regular['frequency'].rank(method='first'),
3     3,
4     labels=[1,2,3]
5 ).astype(int)
```

Monetary

```
1 rfm_regular['M_score'] = pd.qcut(
2     rfm_regular['monetary'].rank(method='first'),
3     3,
4     labels=[1,2,3]
5 ).astype(int)
```

Final RFM Score

```
1 rfm_regular['RFM_score'] = (
2     rfm_regular['R_score'] +
3     rfm_regular['F_score'] +
4     rfm_regular['M_score']
5 )
```

RFM Segmentation

```
1 def segment(row):
2     if row['RFM_score'] >= 8:
3         return 'Loyal User'
4     elif row['RFM_score'] >= 6:
5         return 'Potential Loyalist'
6
7     else:
8         return 'Regular User'
9
10 rfm_regular['segment'] = rfm_regular.apply(segment, axis=1)
11
```

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Appendix - RFM Python Code

combine super user back

```
1 final_rfm = pd.concat([
2     super_users[['user_id', 'recency', 'frequency', 'monetary', 'segment']],
3     rfm_regular
4 ], ignore_index=True)
5
6 final_rfm
```

... [Show hidden output](#)

Next steps:

[Generate code with final_rfm](#)

[New interactive sheet](#)

create table user master

```
1 user_master = df[['user_id', 'user_name']].drop_duplicates()
```

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Appendix - RFM Python Code

join with final_rfm

```
1 final_rfm_named = final_rfm.merge(  
2     user_master,  
3     on='user_id',  
4     how='left'  
5 )
```

```
1 final_rfm_named = final_rfm_named[[  
2     'user_id',  
3     'user_name',  
4     'segment',  
5     'recency',  
6     'frequency',  
7     'monetary',  
8     'R_score',  
9     'F_score',  
10    'M_score',  
11    'RFM_score'  
12 ]]
```

```
1 final_rfm_named
```

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Appendix - RFM Detail score

user_	user_name	segment	recency	frequer	monetary	R_sco	F_sco	M_sco	RFM_sco
user_11	PT Travel Nusantara	Potential Loyalist	1	1,532	356,680.00	3	2	2	7
user_13	PT Saham Global	Potential Loyalist	1	1,516	351,160.00	3	2	2	7
user_16	PT Estate Sejahtera	Potential Loyalist	1	1,566	356,060.00	2	3	2	7
user_18	CV Jalan Mandiri	Potential Loyalist	1	1,555	362,530.00	2	2	3	7
user_4	PT Konektivitas Indo	Potential Loyalist	1	6,152	1,353,130.00	1	3	3	7
user_5	PT Karya Sejahtera	Potential Loyalist	1	6,111	1,385,400.00	1	3	3	7
user_6	PT Logistik Mandiri	Potential Loyalist	1	5,926	1,358,890.00	1	3	3	7
user_7	PT Retail Sejahtera	Potential Loyalist	1	6,048	1,389,150.00	1	3	3	7
user_8	PT Belajar Nusantara	Potential Loyalist	1	6,022	1,346,990.00	1	3	3	7
user_19	CV Niaga Indo	Potential Loyalist	1	1,529	352,210.00	2	2	2	6
user_20	CV Hotel Jaya	Potential Loyalist	1	1,539	362,240.00	2	2	2	6
user_10	PT Mart Jaya	Regular User	1	1,500	338,530.00	3	1	1	5
user_12	CV Mart Indo	Regular User	1	1,466	318,010.00	3	1	1	5
user_14	CV Hotel Nusantara	Regular User	1	1,511	320,780.00	3	1	1	5
user_15	PT Edu Mandiri	Regular User	1	1,475	341,980.00	3	1	1	5
user_17	PT Karya Jaya	Regular User	1	1,495	334,120.00	2	1	1	4
user_9	PT Citra Global	At Risk	1	1,507	332,460.00	1	1	1	3
user_3	CV Wisata Indo	Super User	1	24,307	5,469,540.00				
user_2	PT Dana Nusantara	Super User	1	24,204	5,452,130.00				
user_1	CV Land Sejahtera	Super User	1	24,182	5,400,190.00				

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Appendix - Pitching Video



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